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DL²F: A Deep Learning model for the Local Forecasting of renewable sources

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ABSTRACT

This work presents the first version of DL²F, a powerful Deep Learning model specifically designed to forecast four essential weather variables that influence solar power potential: Global Horizontal Irradiance (GHI), temperature, atmospheric pressure, and relative humidity. The model uses a time series for each variable as input and the forecast provided by the well-known Fifth-Generation Mesoscale Model (MM5) system. Then, it generates new predictions for each variable and the subsequent 3 days. The suggested approach improves the MM5 system's accuracy, effectively mitigating its tendency to misestimate real data in some specific seasons. In essence, DL²F is a regressor able to refine the forecasts generated by the MM5 system and improve its overall precision. The model's architecture is based on a set of recurrent neural networks of the Gated Recurrent Unit (GRU) type.

1. Introduction

Renewable energy sources, such as solar and wind power, are gaining popularity thanks to their clean and sustainable nature (IEA, 2023). However, accurately estimating energy production from these sources is crucial for ensuring a reliable and consistent supply, aiding in the planning and managing the power grid. Predicting their power generation is notably challenging due to their dynamic behaviour, influenced by factors like time, weather parameters and location. Beyond the inherent complexity of these phenomena, the difficulties are further compounded by the limited availability of local real-time data, particularly for short-term forecasts. Therefore, the use of sophisticated models and forecasting techniques is essential to achieve this goal (Harvey, Leybourne, & Whitehouse, 2017).

Precise estimations offer numerous benefits, including reduced greenhouse gas emissions and enhanced energy independence. Without accurate forecasts, the potential advantages of renewable energy sources may be compromised. A wrong estimation of renewable energy production, in fact, may unbalance the distribution network with significant failure in supply. In the case of absent measured data, Regional Climate Models (RCMs) could represent a practical and valuable tool for describing the climatology of places. The Fifth-Generation Mesoscale Model (MM5) is undoubtedly the most adopted among them. Besides the well-known MM5 potentialities, various works point out its limits in predicting weather variables, especially in the case of complex orography, and their tendency to under/overestimate weather parameters (Harrison, 2003; Silva, Meza, & Castellón, 2009). The MM5 capability in forecasting meteorological variables has proven, for example, a broad spectrum of error characteristics on subregional and sub-annual scales (Awan, Truhetz, & Gobiet, 2011). Moreover, a larger influence on model performance and a higher spread are expected for long-term simulations (Done, Leung, Davis, & Kuo, 2005) or during summertime, when the small-scale processes show a more significant influence than in winter (Awan et al., 2011; Fernández, Montávez, Sáenz, González-Rouco, & Zorita, 2007).

A Recurrent Neural Network (RNN) that combines experimental data with MM5 information is proposed to address these challenges and enhance the accuracy of forecasting procedures. This approach integrates machine learning algorithms designed to adapt to new data, optimising accuracy over time by identifying patterns and trends in large datasets. RNN-based models can additionally incorporate multiple sources of information and dynamically adjust predictions based on changing circumstances. Such a method minimises the typical MM5 error in estimating weather parameters and ensures valuable predictions throughout the year.

In detail the research outlines the experimentation process and presents outcomes that validate the effectiveness of this technique in improving the forecasting of meteorological variables. These results offer promising prospects for enhancing reliable and accurate energy-based climate forecasting systems. It is essential to underline that the suggested approach considers multiple weather variables rather than focusing on a single output, as is common. Furthermore, it predicts the investigated outputs by correlating the weather variables.

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Nomenclature

RCM	Regional Climate Model
$MM5$	Fifth-Generation Mesoscale Model
GRU	Gated Recurrent Unit
AI	Artificial Intelligence
DL	Deep Learning
$DLWP$	Deep Learning for Weather Prediction
ML	Machine Learning
ABR	Adaboost Regression
GBR	Gradient-Boosting Regression
RFR	Random Forest Regression
BR	Bagging Regression
$SVMR$	Support Vector Machine Regression
$LSTM$	Long Short-Term Memory
$convLSTM$	Convolutional LSTM
$SARIMA$	Seasonal Autoregressive Integrated Moving Average
ANN	Artificial Neural Network
MLP	Multi-Layer Perceptron
EM	Expectation–Maximisation
MAE	Mean Absolute Error
T	Air temperature [°C]
PSU	Penn State University
$NCAR$	National Center for Atmospheric Research
$SRTM$	Shuttle Radar Topography Mission
DTM	Digital Terrain Model
z_t	Update gates
r_t	Reset gates
t	Time step
W, U	Trainable weight matrices
b	Trainable bias vector
h_{t-1}	Previous hidden state
x_t	Input at time step t
$\sigma(\cdot)$	Sigmoid function
h_t	Current hidden state
$\odot$	Element-wise product operator
h'_t	Candidate activation vector
$S2V$	Sequence to Vector
$BPTT$	Backpropagation Through Time
GHI	Global Horizontal Irradiance [W/m^2]
MAE	Mean Absolute Error
MSE	Mean Squared Error
PBL	Planetary Boundary Layer
LSM	Land Surface Model
MRF	Medium Range Forecast
$GDAS$	Global Data Assimilation System
R^2	Coefficient of Determination
$\hat{y}_i$	Prediction
y_i	True value
N	Number of samples
σ	Standard deviation
R	Correlation coefficient
$RMSD$	Centred root mean square difference
f	Simulated/predicted field
r	Measured field

This paper is structured as follows: after this brief introduction, Section 2 provides an overview of artificial intelligence techniques for forecasting climatic parameters and the investigated weather variables. Section 3 presents background information on MM5 and the adopted Gated Recurrent Unit (GRU) architecture. In Section 4, the proposed model is explained, while Section 5 showcases the obtained results and provides comparisons among the estimated weather datasets. Section 6 analyses the model's performance in non-standard climatic conditions characterised by outlier values. Section 7 provides insights into our solution's deployment and scalability in a production environment. Finally, Section 8 draws conclusions and identifies potential future research directions.

2. Related work

In recent years, significant advancements have been made in leveraging Artificial Intelligence (AI) to predict weather parameters. Researchers and meteorologists have delved into various AI techniques to improve the accuracy and reliability of weather forecasts. These systems can learn complex nonlinear relationships and handle vast amounts of data, enabling them to overcome the limitations of traditional forecasting approaches (Benti, Chaka, & Semie, 2023).

The literature review in this sector underlines the capability of Deep Learning (DL) methods in correctly describing time series problems, particularly when dealing with well-defined spatiotemporal correlations (Ren et al., 2021). Currently, Deep Learning for Weather Prediction (DLWP) strategies can be categorised into three main groups:

- The first category encompasses architectures that rely on simple deep neural network models, like those built upon Autoencoders, Convolutional Neural Networks, and Long Short-Term Memory networks.
- The second category comprises hybrid architectures combining the basic deep neural networks to capture more intricate temporal and spatial problems, entirely driven by the available data.
- The third category consists of coupling architectures that integrate both deep neural networks and numerical models. These architectures are driven by data and guided by theoretical principles to enhance forecasting performance. The proposed approach falls into this category.

The effectiveness of DLWP models in predicting climatic parameters is a relevant topic in the scientific community. Machine learning (ML) and deep learning algorithms have been applied to forecast solar radiation values, resulting in improved performances for both daily and hourly outputs (Alizamir, Kim, Kisi, & Zounemat-Kermani, 2020; Fan et al., 2019; Gürel, Ağbulut, & Biçen, 2020; Huertas-Tato et al., 2020; Khosravi, Koury, Machado, & Pabon, 2018; Srivastava, Tiwari, & Giri, 2019; Voyant et al., 2017). Other works have been centred on investigating the wind source and how to improve its short and long-term estimation through AI models, identifying techniques for maximising the calculated error and enhancing the precision of forecasting (Cadenas, Rivera, Campos-Amezcua, & Heard, 2016; Demolli, Dokuz, Ecemis, & Gokcek, 2019; Ding, Bai, Liu, Fan, & Yang, 2022; Dong et al., 2021; Duan et al., 2021; Li, Zhao, Tseng, & Tan, 2020; Tian, 2020; Wang & Hu, 2015; Xiao, Shao, Jin, & Wu, 2021; Zendehboudi, Baseer, & Saidur, 2018). Additionally, various analyses have evaluated topics unrelated to climatic parameters for energy production, such as predicting daily air temperature, future rainfall, and tropical cyclone intensities (Roy, 2020; Shi et al., 2015; Yang, Lee, & Tippett, 2020).

These findings demonstrate the remarkable effectiveness of AI techniques in the field. However, many existing studies have focused on investigating a single variable at a time. In contrast, this research aims to leverage the power of data-driven models to analyse a broad range of climatic factors simultaneously. This approach provides a more holistic understanding of the complex interactions and dynamics within the climate system, contributing to improved climate prediction and mitigation efforts. The investigated weather variables are briefly reported in the following subsections.

2.1. Solar radiation prediction

Solar radiation forecasting is crucial for the efficient planning and operation of solar energy systems. Accurate predictions enable power grid operators and energy traders to make informed electricity production and distribution decisions. Recently, machine learning algorithms have emerged as promising tools to enhance solar radiation forecasting accuracy.

Several works have explored machine-learning techniques for forecasting solar radiation. For instance, one study (Mohandes, Rehman, & Halawani, 1998) introduces a neural network approach to estimate global solar radiation in Saudi Arabia. The research uses data from 41 radiation collection stations across the kingdom, with 31 locations adopted for training the neural networks and 10 locations for testing. Results show that this technique is viable for spatial modelling of solar radiation.

In another work (Alam, Al-Ismail, Hossain, & Rahman, 2023), researchers focus on ensemble methods for solar radiation forecasting, which combine multiple models to improve accuracy. They use meteorological data from 32 stations, including variables such as maximum temperature, minimum temperature, total rain, humidity, sunshine, wind speed, cloud coverage, and irradiance. The study develops ensemble machine learning algorithms like AdaBoost Regression (ABR), Gradient-Boosting Regression (GBR), Random Forest Regression (RFR), and Bagging Regression (BR) to predict solar irradiance.

Overall, these related works (Mellit & Kalogirou, 2008; Yadav & Chandel, 2014) demonstrate the potential of machine learning techniques in improving solar radiation forecasting. However, further research is needed to develop more reliable and scalable models that handle diverse weather and climate conditions. Advancements in this field hold promise for optimising solar energy utilisation and sustainability.

2.2. Temperature prediction

Weather prediction plays an essential role in our daily lives, as it helps us to anticipate different weather conditions and plan our schedules accordingly. Machine learning algorithms have demonstrated remarkable potential in refining the precision of meteorological forecasting, particularly in estimating temperatures. Many studies have explored ML techniques for temperature prediction, such as neural networks and support vector regression. Integrating these advanced ML algorithms into weather forecasting systems can significantly enhance the accuracy of temperature predictions, benefiting various industries like agriculture and transportation.

For instance, Paniagua-Tineo et al. (2011) presents a technique using Support Vector Machine Regression (SVMR) to forecast the highest temperature of each day. The accuracy of SVMR's predictions is demonstrated by considering various meteorological factors and comparing them with other types of neural networks.

In another study (Kreuzer, Munz, & Schlüter, 2020), the improvement of local temperature predictions up to 24 h ahead is examined using deep neural networks. The study employs univariate long short-term memory (LSTM) networks and 2D-convolutional LSTM (convLSTM) networks, comparing them with the Seasonal Autoregressive Integrated Moving Average (SARIMA) model. Based on a case study conducted using data from five weather stations in Germany, both multivariate LSTM and convolutional LSTM models exhibited superior performance compared to the other tested approaches.

2.3. Atmospheric pressure prediction

The significance of atmospheric pressure as a crucial weather factor affecting our daily routine cannot be overstated. Precise forecasts of atmospheric pressure could help us to predict cloudy days that will influence solar radiation or also to prepare for severe weather occurrences like hurricanes and storms. As a result, several research endeavours have explored the use of machine learning techniques to predict atmospheric pressure.

This research (Ray, Mukhopadhyay, Datta, & Pal, 2013) presents a new Artificial Neural Network (ANN) model that incorporates previous measurements of weather-related variables, such as temperature, relative humidity, air pressure and vapour pressure during the training process. This comprises a Multi-Layer Perceptron network (MLP) combined with a hybrid MLP and k-means clustering method to yield more accurate predictions.

Similarly, another study (Kim et al., 2016) introduces an approach that employs machine learning techniques to enhance atmospheric pressure data correction acquired through smartphones. The proposed method involves employing time domain classification in combination with clustering and regression analysis. The findings indicate that leveraging Expectation-Maximisation (EM) clustering for regression analysis led to a 26% reduction in mean absolute error while machine learning models outperformed traditional linear regression methods regarding lower Mean Absolute Error (MAE) values. Overall, this novel technique is promising in advancing atmospheric pressure readings' predictive accuracy and efficiency.

2.4. Relative humidity prediction

The significance of forecasting relative humidity in weather predictions should not be ignored. Accurate predictions of relative humidity offer valuable knowledge for several practical purposes, including agriculture, air quality monitoring, and planning outdoor activities. Recently, numerous studies have investigated using artificial neural networks to forecast relative humidity. These analyses have demonstrated promising results, highlighting the potential effectiveness of artificial neural networks in predicting this climatic parameter with high accuracy (Ghadiri, Marjani, Mohammadinia, & Shirazian, 2021).

Similarly, another study conducted by Hanoon et al. (2021) suggests various machine learning algorithms for predicting air temperature (T) and relative humidity. These algorithms include gradient boosting trees, random forests, and linear regressions, as well as different artificial neural network architectures like multi-layered perceptron and radial basis function. These methods are effective in predicting relative humidity.

All these studies affirm the successful application of artificial neural networks and machine learning methodologies in predicting relative humidity. By assimilating historical meteorological data and employing advanced algorithms, these techniques enhance precision and efficiency in forecasting relative humidity levels.

3. Background

This section reports information and concepts related to the MM5 system and the GRU neural networks.

3.1. The MM5 system

The Fifth-Generation Mesoscale Model, developed by the Penn State University (PSU) and the National Center for Atmospheric Research (NCAR), is a powerful numerical weather prediction model capable of simulating and forecasting mesoscale and regional-scale atmospheric circulation. Its portability to various computer platforms has made MM5 widely used for diverse applications, including short-term weather and air quality forecasts and climate modelling.

This non-hydrostatic model employs complex mathematical equations governing thermo and fluid dynamic mechanisms. Specifically, MM5 is based on momentum and energy conservation equations and adopts a tendency equation for the perturbation pressure (model prognostic on the pressure variable). It uses a σ pressure coordinate system and finite difference numerical schemes (Arakawa-Lamb B-staggering) for predicting a wide range of meteorological variables such as temperature, atmospheric pressure, wind speed and direction, solar radiation, and cloud cover. For more detailed information, please refer to Grell, Dudhia, and Stauffer (1994) and Dudhia et al. (2005).

To better describe the site's climatology and improve the model's accuracy, a modified version of MM5 is used; indeed, the Fortran code of the standard version was opportunely modified to ingest the 3 arc-second Shuttle Radar Topography Mission (SRTM) Digital Terrain Model (DTM) data and the CORINE European land cover database (European Environment Agency (EEA), 2023). This allows the simulation of weather parameters with higher resolutions up to 0.2 km and to describe phenomena influenced by specific local terrain conformation.

Moreover, a 2-way nesting procedure with 5 different domains is employed to predict all meteorological phenomena across local to synoptic scales. The domains are chosen by always setting a nesting ratio equal to 3:1, and they are all centred on the same point, corresponding to the interested area, with the same xy grid points (31×31).

3.2. Recurrent Neural Networks

Recurrent Neural Networks are a class of artificial neural networks designed explicitly for sequence data, making them particularly valuable in tasks like natural language processing and time series analysis. Two popular variants of RNNs are long short-term memory and gated recurrent unit. While both LSTM and GRU effectively capture sequential dependencies, GRUs are notably simpler in architecture. This simplicity often results in shorter training times and less memory consumption.

In the present study, we selected GRU over LSTM after directly comparing both models on a subset of cases. In this comparative analysis, we found that the performance of the models was quite similar. As a result, we decided to adopt the GRU due to its simpler architecture.

3.3. GRU neural networks

The GRU architecture employs a selective gating mechanism to capture long-term dependencies within input sequences (Cho et al., 2014). This architecture is built upon GRU cells designed to retain or discard information selectively. They achieve this by updating a hidden state based on the current input.

At each time step, every cell processes the previous hidden state and the current input to calculate the current hidden state. Furthermore, this hidden state is also utilised to compute the output.

This selective gating mechanism enables the GRU to capture long-term dependencies in the input sequence, mitigating the vanishing gradients problem (Hochreiter & Schmidhuber, 1997).

The core components of this architecture are the *update gate* and the *reset gate*. The update gate controls how much of the previous hidden state should be preserved and how much new input should be incorporated into the current hidden state. On the other hand, the reset gate controls how much of the previous hidden state should be forgotten.

Formally, the update and reset gates can be defined as follows:

$$z_t = \sigma(W_z \cdot x_t + U_z \cdot h_{t-1} + b_z) \tag{1}$$

$$r_t = \sigma(W_r \cdot x_t + U_r \cdot h_{t-1} + b_r) \tag{2}$$

where z_t and r_t are the update and reset gates at time step t, W_z, U_z, W_r and U_r are *trainable weight matrices*, b_z and b_r are *trainable bias vectors*, h_{t-1} is the previous hidden state, x_t is the input at time step t, and $\sigma(\cdot)$ is the *sigmoid* function. The current hidden state, h_t, is then computed as follows:

$$h'_t = tanh(W \cdot x_t + U \cdot (r_t \odot h_{t-1}) + b) \tag{3}$$

$$h_t = z_t \odot h_{t-1} + (1 - z_t) \odot h'_t \tag{4}$$

where W and U are *trainable weight matrices*, b is a *trainable bias vector*, and $\odot$ is the *element-wise product* operator.

The vector h'_t, called *candidate activation vector*, is calculated using the current input x_t and the previous hidden state h_{t-1}. Conceptually, it represents a proposal for the new hidden state. The GRU subsequently determines whether to accept this proposal or partially reject it based on the update gate z_t.

Specifically, the resulting hidden state h_t is obtained by adding together h_{t-1} and h'_t, weighted by the factors z_t and $(1 - z_t)$. This means that the update gate z_t controls how much of the previous hidden state should be retained (weighted by z_t), and how much of the candidate activation vector h'_t should be incorporated (weighted by $(1 - z_t)$). Fig. 1(a) reports the structure of a GRU cell.

It is possible to design GRU models with multiple layers to capture more complex patterns in the input sequences. In this case, the output of one layer, i.e. the hidden state of the related GRU cell, is the input for the next layer (as shown in Fig. 1(b)).

We adopt an *S2V* (*Sequence to Vector*) architecture, which calculates the output by processing the hidden state of the last layer in the final time step, with a *fully connected layer* (Fig. 1(c)).

To train a GRU neural network, we need to define a suitable *loss function* and use *Backpropagation Through Time* (*BPTT*) algorithm (Werbos, 1988) to optimise its parameters.

As we will see in the next section, the core of the proposed architecture is a set of multi-layer GRU models.

4. Basic framework

The model we present is a regressor to refine the forecasts of the MM5 system, making them more accurate.

4.1. Problem formulation

A typical weather station measures the following six physical variables:

- *Global Horizontal Irradiance* (*GHI*)
- *Temperature*
- *Atmospheric Pressure*
- *Relative Humidity*
- *Wind Speed*
- *Wind Direction*

In this work, we have considered two meteo-masts equipped with facilities for solar irradiation analyses. We choose a subset of k physical variables for each geographical location from the list mentioned above. In both case studies, we exclude the variables "*wind speed*" and "*wind direction*" because the measurements for these two parameters tend to be highly dynamic and noisy, considering the location of the meteorological towers too close to large obstacles that produce significant wakes. Additionally, in the second case study, we exclude the "*temperature*" variable as it is not detectable by the weather station. As a result, the present research has $k = 4$ in the first case and $k = 3$ in the second one.

Since the physical variables we are analysing exhibit seasonal trends, we also incorporate two *time variables*: the "*day of the year*" and the "*hour of the day*" corresponding to each sample.

The values of each variable are normalised.

To select the most suitable normalisation technique, we analysed the data distribution of each weather parameter. These distributions

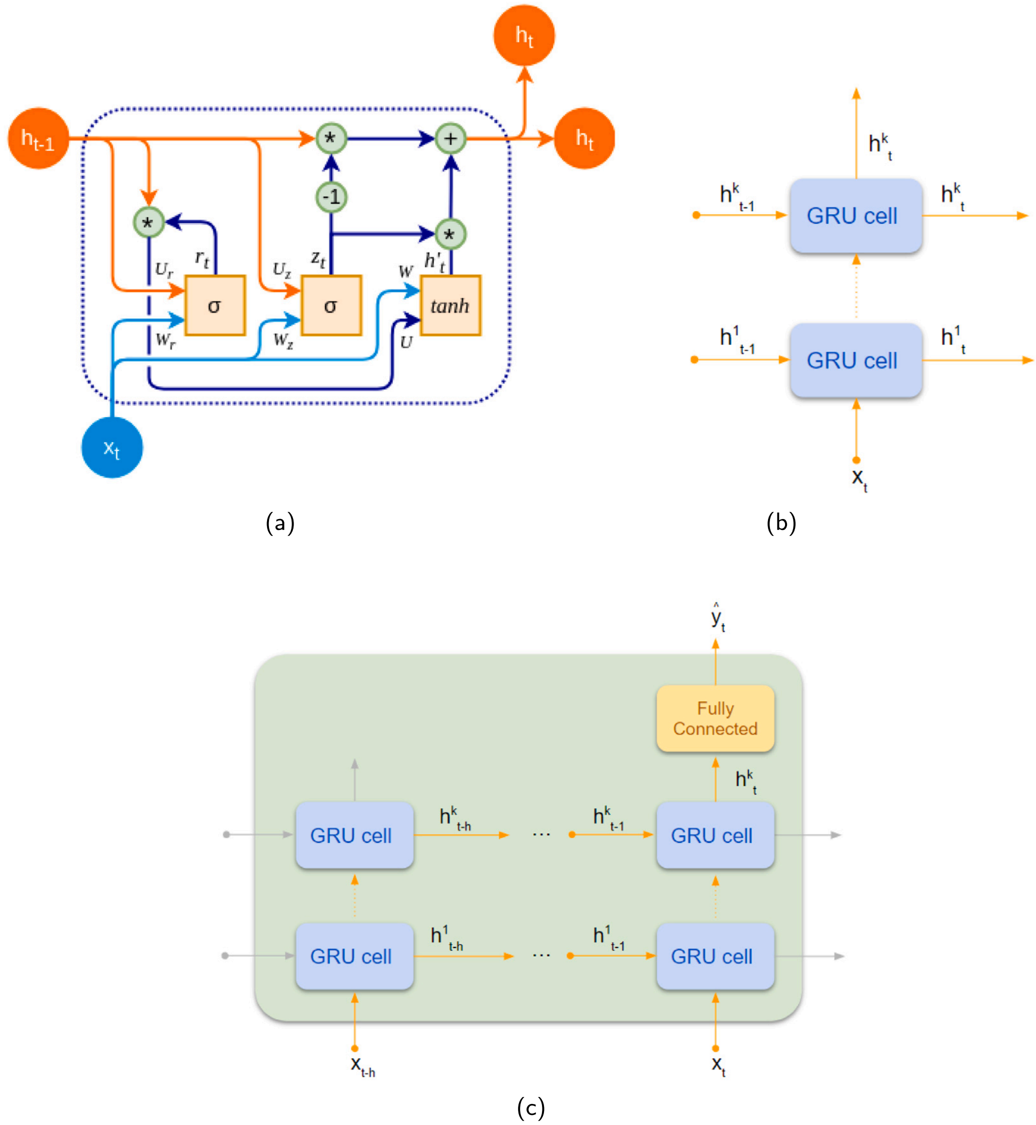


Fig. 1. Schematisation of the adopted GRU cell (a), the K-layer GRU (b) and the S2V GRU model (c).

significantly deviate from a *normal distribution*, displaying distinct skewed patterns. *Standard normalisation* (*Z-score normalisation*) tends to perform better when the original data distribution is close to a normal distribution. Therefore, we decided to adopt *min–max normalisation*.

The normalised real values detected by the weather station at time t are stored in a data point $x_t^r \in \mathbb{R}^k$ in the same order reported earlier. Similarly, the MM5 system provides the corresponding normalised values for the same location, stored in a data point $x_t^m \in \mathbb{R}^k$.

Since the MM5 system is capable of forecasting the variables we are considering, it can provide the data point $x_{t+\tau}^m \in \mathbb{R}^k$ at time t for a given *time delay* $\tau \in \mathbb{N}$.

Finally, the values related to the time variables are stored in a data point $x_t^d \in \mathbb{R}^2$.

Therefore, at time t, the data point $x_t = [x_t^d, x_t^r, x_{t+\tau}^m] \in \mathbb{R}^{2k+2}$ is available. As the input values are normalised, $x_t[i] \in [0,\ 1]$, for each $0 \leq i \leq (2 \cdot k + 1)$.

Given a time delay $\tau \in \mathbb{N}$, $h \in \mathbb{N}_0$, and the time series $X = \{x_t \mid t \in \mathbb{N}_0\}$, our goal is to forecast an estimate $\hat{y}_t = x_{t+\tau}^p$ that approximates $y_t = x_{t+\tau}^r$ better than $x_{t+\tau}^m$, for each $t \geq h$, with respect to a given error function (e.g. *Mean Absolute Error*).

4.2. Architecture

As previously reported, each input value is scaled by a *min–max normaliser*.

To compute $x_{t+\tau}^p$, for each $t \geq h$, we designed a set of k GRU neural networks $\{M_0, \ldots, M_{k-1}\}$.

In detail, for each $i \in [0\ ..\ k-1]$, at time t the model M_i processes the sequence $\tilde{x}_t = [x_{t-h}, \ldots, x_t]$, and returns $\hat{y}_t = x_{t+\tau}^p[i]$.

The *hyperparameters* of M_i, such as the *number of hidden layers* and *dimension of each hidden layer*, will be tuned to maximise its performance, with respect to the physical variable it forecasts.

The *fully-connected* network that returns the model's output (see Fig. 1(c)) has one hidden layer with a number of neurons equal to half the size of the hidden state. The neurons of this layer have a *ReLu* activation function. The output layer consists of a single neuron with no activation function.

The outputs corresponding to the "*GHI*" variable undergo additional refinement through a simple post-processing module. This module takes $x_{t+\tau}^p[0]$ as input and returns the same value if $t+\tau$ is a daytime instant, and 0 if it is a nighttime instant.

The system's architecture is illustrated in Fig. 2.

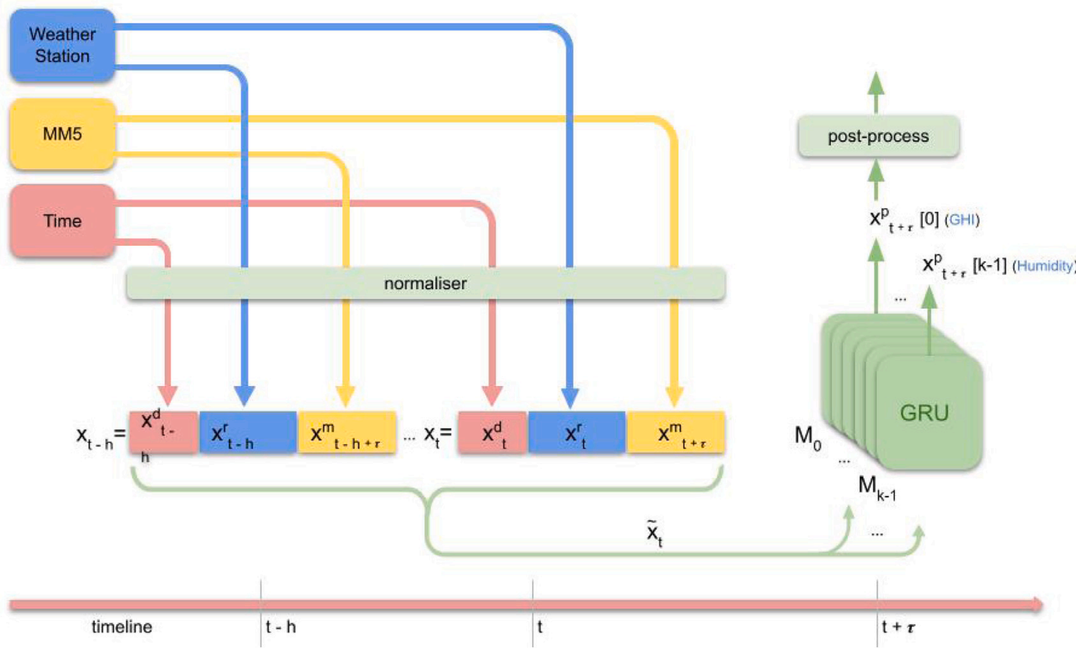


Fig. 2. Model architecture.

It is essential to emphasise that specific models must be built and trained for each geographical site because every site has unique properties (e.g. terrain conformation, exposure to sunlight, proximity to rivers/sea, etc.) that significantly influence the trend of the physical quantities under analysis.

4.3. Training

Given a geographical location and a variable $i \in [0 \,..\, k-1]$, the time series $X = \{x_t \mid t \in \mathbb{N}_0\}$, the corresponding set of sequences $\tilde{X} = \{\tilde{x}_t \mid t \geq h\}$, and the related labels $Y = \{y_t \mid t \geq h\}$, where $y_t = x^r[i]_{t+\tau}$, are available.

The sets $\tilde{X}$ and Y are split into three consecutive segments: (i) *training set* (60%); (ii) *validation set* (20%), and (iii) *test set* (20%).

To train M_i, we perform a *batch gradient descent* process that tunes its parameters in order to minimise the error on its predictions for each *batch* of data concerning a *loss function*. The loss function we have selected is the Mean Absolute Error (MAE).

Therefore, for a given t, the error on a single batch containing B samples $\{\tilde{x}_{t+j} \mid j \in [0 \,..\, B-1]\}$, and the corresponding labels $\{y_{t+j} \mid j \in [0 \,..\, B-1]\}$, where $y_{t+j} = x^r_{t+\tau+j}[i]$, is computed as:

$$\mathrm{E} = \frac{1}{B} \sum_{j=0}^{B-1} |x^p_{t+\tau+j}[i] - x^r_{t+\tau+j}[i]| \tag{5}$$

The selection of the MAE loss function over other options, such as Mean Squared Error (MSE) or Huber Loss, was carefully considering our forecasting problem's specific requirements.

The main reason for choosing MAE as the loss function is its robustness to outliers. In our forecasting task, we are dealing with weather data and variables that can be subject to unexpected and extreme fluctuations due to various factors, such as sudden weather events. For instance, GHI is influenced by various environmental factors, including cloud cover, which can introduce sporadic and significant fluctuations in the data. These fluctuations can result in extreme outliers that do not necessarily reflect the typical behaviour of the system. Similarly, temperature, atmospheric pressure, and relative humidity can exhibit rapid and extreme variations due to several weather events.

MSE, which squares errors, can heavily penalise large outliers, making the model more sensitive to extreme values in the data. This can result in sub-optimal model performance because, in an attempt to minimise the high outliers' squared errors, the accuracy of standard predictions is reduced. Instead, MAE ensures that the model's performance remains resilient despite outliers.

While the Huber Loss is another option combining the advantages of MAE and MSE by using a parameter to tune the transition between the two, we found that, in our case, the straightforward nature of MAE aligns well with our objectives.

The training process is performed for multiple *epochs*. To prevent the *overfitting* problem, the *early stopping* technique is adopted.

Early stopping uses a *validation set*, and the decision to stop training is based on the model's performance on this set. The technique monitors the *validation loss*, and training is halted when the loss fails to decrease for a certain number of epochs (known as *patience*).

We carefully tuned this hyperparameter to address the problem of potential *underfitting*. Setting an appropriate patience value ensures that the model is given sufficient time to learn from the data and generalise effectively. We performed multiple experiments to find a balance that prevents overfitting without causing underfitting. A simple flow-chart describing the training process is reported in Fig. 3(a).

4.4. Inference

After training each model M_i, for each $i \in [0 \,..\, k-1]$, the system is ready to generate new predictions. For every time step t, the system normalises the sequences of real and MM5 values, resulting in $[x^r_{t-h}, \ldots, x^r_t]$ and $[x^m_{t-h+\tau}, \ldots, x^m_{t+\tau}]$ respectively. The system calculates $\tilde{x}_t$ using these sequences and the time-steps sequence. This transformed input is then fed into each model M_i, for each $i \in [0 \,..\, k-1]$, which computes the value $\hat{y}_t = x^p_{t+\tau}[i]$. In the final step, the system denormalises $\hat{y}_t$ to obtain the final forecast.

We reported a simple flowchart describing the inference process in Fig. 3(b).

5. Evaluation

In this study, we used the MM5 model to predict climatic parameters for two Italian cities: Ottana on Sardinia Island (40°14′03″ northern

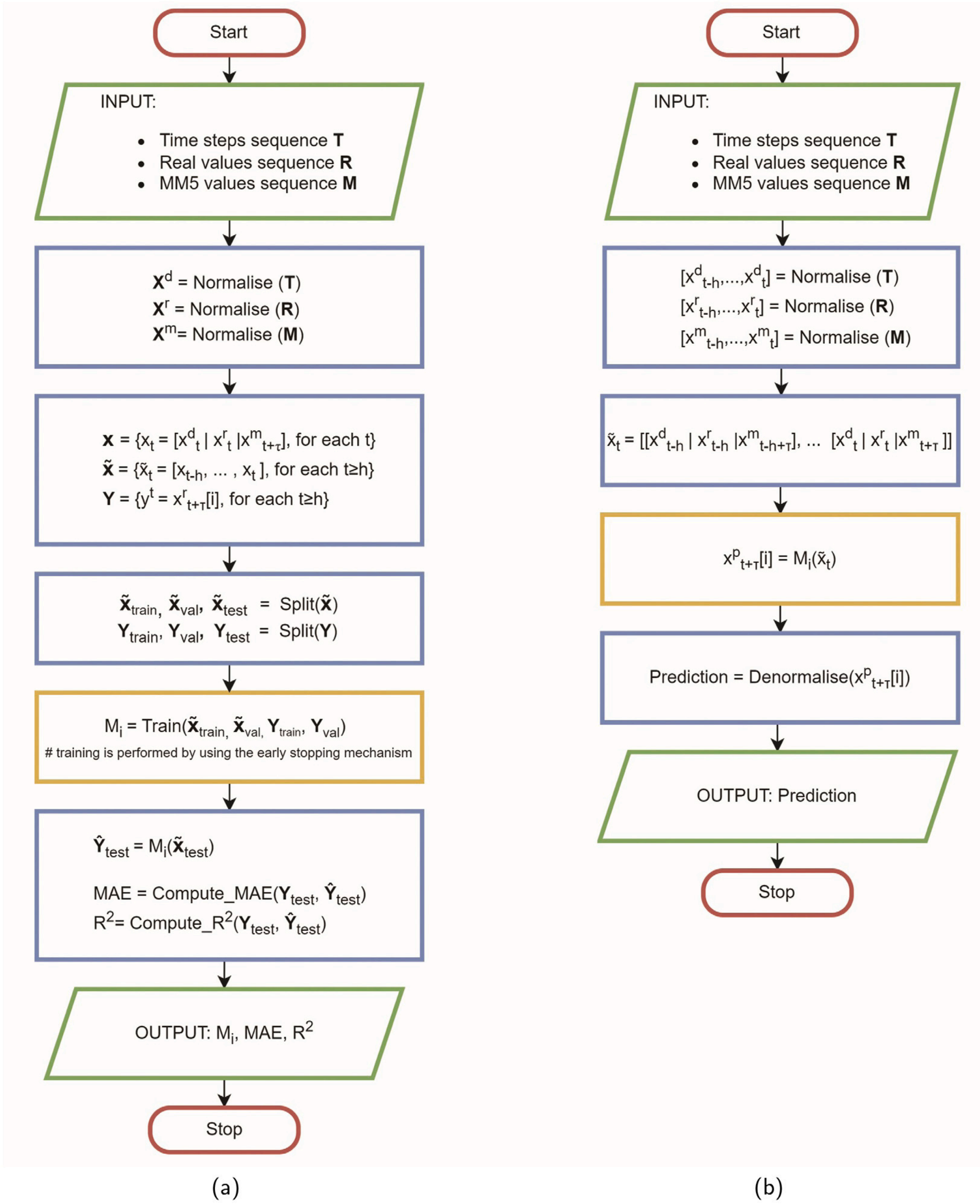


Fig. 3. Flowcharts of the training (a) and inference (b) processes.

latitude and 09°32′33″ eastern longitude) and Casaccia in the Lazio Region (42°02′30″ northern latitude and 12°18′15″ eastern longitude).

For our simulations, we selected the Planetary Boundary Layer (PBL) and the Land Surface Model (LSM) that correspond, respectively, to the Medium Range Forecast model (MRF) developed by Hong and Pan (1996) and the NOAH LSM. The choice of the MRF model combined with the NOAH LSM comes from the major findings obtained from previous work (Silvero, Lops, Montelpare, & Rodrigues, 2019).

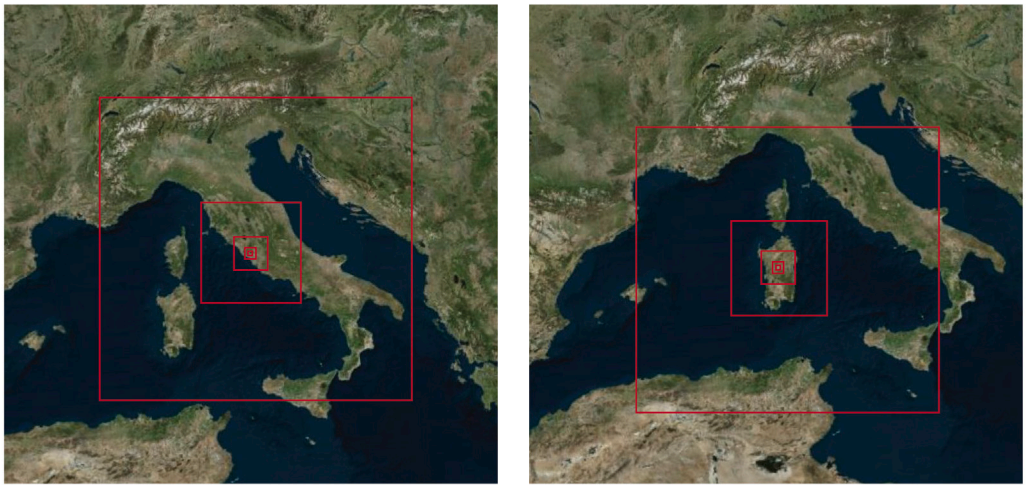
The modelling setup involves five 2-way nesting domains, ranging from a node-to-node distance of 0.4 km covering an area of 144 km^2 for the innermost domain to a grid spacing of 32.4 km covering an extended area of 944784 km^2 for the outermost domain. Additionally, we implemented 26 sigma levels for the vertical resolution.

For visual reference, Fig. 4 illustrates the nested domain for Casaccia (Fig. 4(a)) and Ottana (Fig. 4(b)), while Table 1 provides a summary of the main characteristics of the domains.

Table 1
Main features of the mesoscale domains for the Ottana and Casaccia virtual anemometers.

Domain	Grid spacing [km]	Grid points	Domain extension [km^2]
D5	0.4	31 × 31	12 × 12
D4	1.2	31 × 31	36 × 36
D3	3.6	31 × 31	108 × 108
D2	10.8	31 × 31	324 × 324
D1	32.4	31 × 31	972 × 972

The numerical meteorological masts for the analysed locations were determined through hindcasting simulations covering different periods based on the available measured data. For the Ottana weather station, the virtual anemometer estimated the outputs from May 31st, 2021, to May 31st, 2022, while for Casaccia, the covered period extended from January 1st, 2018, to December 31st, 2020.



(a) (b)

Fig. 4. The MM5 nesting domains for Casaccia (a) and Ottana (b).

Table 2
Synthesis of the adopted hyperparameters.

Case study	Variable	Hidden layer dim.	# of hidden layers	Batch size	Learn. rate
Casaccia	GHI	128	2	512	10^{-3}
	Temperature	128	2	512	10^{-4}
	Atmospheric pressure	128	2	512	10^{-4}
	Relative humidity	128	2	512	10^{-4}
Ottana	GHI	256	2	512	10^{-4}
	Atmospheric pressure	64	4	32	10^{-5}
	Relative humidity	128	2	512	10^{-3}

The selected input weather dataset is the NCEP ds083.2 Final Operational Global Analysis data, obtained from the Global Data Assimilation System (GDAS), available on a 1-degree by 1-degree grid. The procedure generates output values at 4-minute intervals and 26 measurement heights, ranging from 10 m to various thousand meters. The resulting database contains physical quantities such as PBL height, Downward Shortwave Radiation, Downward Longwave Radiation, Temperature @ 2 m above surface level a.s.l., wind speed @ measurement plan height a.s.l., wind direction @ measurement plan height a.s.l., temperature @ measurement plan height a.s.l., air pressure @ measurement plan height a.s.l., and relative humidity @ measurement plan height a.s.l.

In our experiments, a one-time step corresponds to 10 min, and we set $h = 432$ (3 days). We conducted three sets of experiments with different values for τ: (i) $\tau = 144$ (1 day), (ii) $\tau = 288$ (2 days), and (iii) $\tau = 432$ (3 days).

Additionally, we tested the model on two different geographical locations. Therefore, we trained and checked 12 models for the first case study and 9 models for the second one.

In each case study, we empirically determined the optimal hyperparameters for each physical variable, ensuring excellent performance of the models. The specific hyperparameters found for each variable are listed in Table 2.

5.1. Datasets

The dataset for both case studies, Casaccia and Ottana, includes a series of measurements sampled at 10-minute intervals. In the Casaccia case study, the measurements were collected over a 3-year period starting from January 1st, 2018, at 1:10 am. Each measurement in this dataset includes the *date*, *time*, and values of four physical quantities ($k = 4$): *GHI*, *temperature*, *atmospheric pressure*, and *relative humidity*, along with their corresponding values from the MM5 system. In the Ottana case study, the measurements were taken over a 1-year duration starting from May 31st, 2021, at 17:10. However, the Ottana dataset includes three physical quantities ($k = 3$): *GHI*, *atmospheric pressure*, and *relative humidity*, along with their respective values from the MM5 system.

All values in each dataset have been normalised. For every selected value, a specific dataset was created for the parameter τ – 144 (1 day), 288 (2 days), and 432 (3 days) – and for each possible variable i – 0 (*GHI*), 1 (*temperature*, only for Casaccia), 2 (*atmospheric pressure*), 3 (*relative humidity*). It contains sequences of the form $\tilde{x}_t = [x_{t-h}, \dots, x_t]$, where $h = 432$ (3 days) and $x_v = [x^d_v, x^r_v, x^m_{v+\tau}]$, for $v \in [t-h \mathrel{..} t]$, along with the corresponding labels $x^r_{t+\tau}[i]$.

Each of these datasets has been divided into three consecutive subsets: (i) *training set* (60%), (ii) *validation set* (20%), and (iii) *test set* (20%).

5.2. Experimental settings

The proposed model was implemented in *Python* (version *3.9.16*) using *PyTorch* (version *1.13.1+cu116*). All experiments were conducted on a machine equipped with an *Intel i9* processor, 32 GB of RAM, and a *GeForce RTX 2080* GPU with 8 GB of VRAM.

The models were trained using the *RMSprop* optimiser. We selected RMSprop over other optimisers with comparable performance for its efficiency in terms of memory usage. In fact, it maintains a moving average of squared gradients for each parameter, which requires less memory than the mechanisms of other optimisers. For instance, Adam maintains both first and second-moment estimates.

The maximum number of *epochs* has been set to 200 and the *patience* parameter of the *early stop process* to 10.

For the sake of brevity, we present only the graph reporting the *train loss* and the *validation loss* of a single model, specifically the one related with Casaccia, the *pressure* variable, and a *delay time* of 3 days (Fig. 5). The graph reports both the original values and a moving average computed with a three-step sliding window for both loss metrics. It clearly shows that during the initial stages of training,

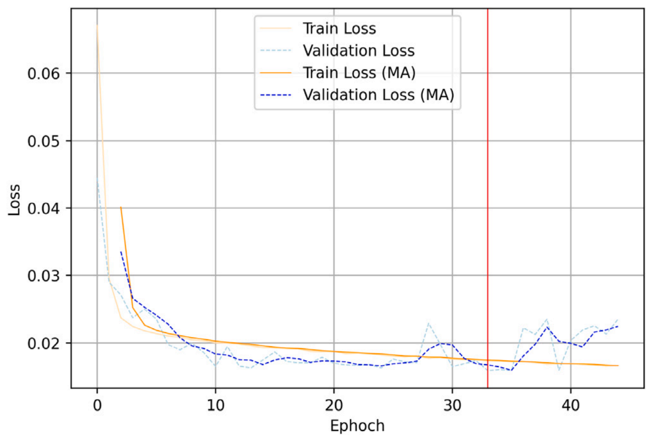


Fig. 5. Train and validation loss of the Casaccia pressure forecasting model (3-day prediction).

both the training loss and validation loss exhibit a decreasing trend. Then, while the training loss decreases, the validation loss increases after the 33rd epoch. The training process only concludes when a new minimum value of the validation loss has not been reached for 10 epochs (the *patience* value). The final model corresponds to the epoch at which the last minimum of the validation loss was detected, marked in the graph by a vertical red line.

In general, for each hyper-parameter, we selected the optimal value by performing a *grid search*.

5.3. Results

The effectiveness of the suggested procedure is assessed using various statistical measures. Specifically, we calculate the MAE and the Coefficient of Determination (R^2), and we also employ Taylor diagrams for the comparisons.

The mean absolute error is a metric used to evaluate the accuracy of a prediction or estimation model. It quantifies the average difference between the predicted and actual values, providing insight into the model's performance in terms of absolute error. The MAE helps understand the average magnitude of the errors, regardless of their direction. It can be defined as shown in Eq. (6), where $\hat{y}_i$ represents the prediction, y_i is the true value, and N is the number of samples.

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|y_i - \hat{y}_i\right| \tag{6}$$

The Coefficient of Determination, on the other hand, represents the proportion of the variance that can be predicted from the independent variables. In simpler terms, it gauges how well the regression model explains the variability of the data points around the mean, thus assessing the goodness of fit of the regression model. The Coefficient of Determination ranges from 0 to 1, with higher values associated with models that better explain the variability in the dependent variable. We adopt Eq. (7) for calculating R^2, where $\bar{y}$ denotes the mean of the observed data.

$$\mathrm{R}^2 = 1 - \frac{\sum_{i=1}^{N}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{N}\left(y_i - \bar{y}\right)^2} \tag{7}$$

The results presented in Table 3 indicate that for the Casaccia study, $\mathrm{DL^2F}$ outperforms the MM5 systems in meteorological forecasting. It demonstrates promising performance across all evaluated metrics, particularly in GHI forecasting, where it achieves superior results compared to MM5 for all selected configurations. Notably, for the 3-day forecasting horizon, our model shows a remarkable 16.19% reduction in errors compared to MM5.

The use of artificial neural networks also underlines the potential for improving temperature forecasts. The neural network models consistently outperform the MM5 systems in temperature forecasting across different periods. The most significant improvement is observed for the 1-day forecast horizon, where the neural network reduces prediction errors by 46.35%.

Considering atmospheric pressure forecasting, ANNs have shown their effectiveness. The MAE error is significantly lower in the neural network models than in MM5, indicating higher accuracy.

Also, the neural network models demonstrated better performance for relative humidity prediction than MM5 systems. Specifically, the neural network models consistently achieved lower MAE error values for this parameter across all selected configurations.

Table 4 presents the results of the Ottana case study. Here, we observe that while GHI and relative humidity forecasting using neural networks outperform traditional MM5 systems, the performance for atmospheric pressure prediction is not as strong as in the Casaccia case study. Nonetheless, $\mathrm{DL^2F}$ still demonstrates very good performance, and the results are comparable.

Figs. 6–8 further illustrate the capability of artificial neural networks to adapt to changing environments. The figures depict real data, MM5 predictions, and neural network predictions for GHI, atmospheric pressure, temperature, and relative humidity variables. Overall, the neural network forecasts closely align with the real data in the Casaccia picture, showcasing their ability to capture and predict patterns accurately.

In the Ottana case study, the neural network forecasts for GHI and relative humidity closely match the real data, further supporting their efficacy in capturing and predicting these meteorological variables. However, for atmospheric pressure, the predicted data sometimes deviate slightly above the real data, indicating that there might be further improvements in pressure forecasting using artificial neural networks.

In conclusion, the results demonstrate that artificial neural networks have the potential to improve accuracy and reduce errors in meteorological forecasting when compared to other systems. Their ability to capture and predict patterns effectively highlights their efficacy in handling such forecasting tasks.

In addition to the comparisons using R^2 and MAE, graphical tools like Taylor diagrams are employed to assess the effectiveness of the adopted approach in representing multivariate data. These diagrams can display multiple variables simultaneously by mapping their Standard Deviation (σ), Correlation Coefficient (R), and Centred Root Mean Square Difference (RMSD) onto a polar coordinate system. In the diagram, the observed values, which serve as the reference data, are placed at the origin, while the simulated values are represented by markers at various distances and angles from the origin.

In detail, the standard deviation quantifies the amount of dispersion or spread in a dataset, estimating the error fluctuations around the dataset's mean. The correlation coefficient measures the strength or consistency between two variables, with values ranging from −1 to +1. A zero value indicates no correlation between two variables; positive values indicate variation in the same direction; negative values indicate variation in opposite directions. Lastly, the root mean square deviation quantifies the two sets' average distance between corresponding data points. It is commonly used to calculate the difference or similarity between two data points, approaching zero when the variables are more alike. For a simulated/predicted field (f) and a measured field (r), the formulas used to calculate these parameters are reported below (Taylor, 2005):

$$R = \frac{\frac{1}{N}\sum_{i=1}^{N}\left(f_i - \bar{f}\right)\left(r_i - \bar{r}\right)}{\sigma_f \cdot \sigma_r} \tag{8}$$

$$RMSD = \frac{1}{N}\sum_{i=1}^{N}\left[\left(f_i - \bar{f}\right) - \left(r_i - \bar{r}\right)\right]^2 \tag{9}$$

$$\sigma_f^2 = \frac{1}{N}\sum_{i=1}^{N}\left(f_i - \bar{f}\right)^2 \tag{10}$$

Table 3
Performance metrics for Casaccia.

Variable	System	$\tau = 144$ (1 day)		$\tau = 288$ (2 days)		$\tau = 432$ (3 days)	
		MAE	R^2	MAE	R^2	MAE	R^2
GHI	MM5	46.557	0.867	46.472	0.863	45.365	0.867
	DL^{2}F	**39.588**	**0.903**	**39.230**	**0.895**	**38.021**	**0.898**
	Δ	−14.97%	4.11%	−15.58%	3.83%	−16.19%	3.58%
Temperature	MM5	2.388	0.841	2.388	0.840	2.361	0.844
	DL^{2}F	**1.281**	**0.944**	**1.398**	**0.937**	**1.299**	**0.943**
	Δ	−46.35%	12.28%	−41.44%	11.50%	−44.99%	11.76%
Pressure	MM5	173.116	0.863	173.264	0.857	170.598	0.867
	DL^{2}F	**85.310**	**0.958**	**95.273**	**0.945**	**87.135**	**0.955**
	Δ	−50.72%	11.01%	−45.01%	10.26%	−48.92%	10.16%
Humidity	MM5	10.720	0.571	10.735	0.570	10.715	0.581
	DL^{2}F	**8.163**	**0.742**	**8.591**	**0.719**	**8.456**	**0.729**
	Δ	−23.86%	29.97%	−19.97	26.24%	−21.09%	25.46%

Table 4
Performance metrics for Ottana.

Variable	System	$\tau = 144$ (1 day)		$\tau = 288$ (2 days)		$\tau = 432$ (3 days)	
		MAE	R^2	MAE	R^2	MAE	R^2
GHI	MM5	60.230	0.846	59.837	0.844	57.498	0.856
	DL^{2}F	**51.270**	**0.890**	**54.071**	**0.882**	**53.639**	**0.875**
	Δ	−14.88%	5.21%	−9.64%	4.55%	−6.71%	2.30%
Pressure	MM5	**60.757**	**0.977**	**64.548**	**0.975**	**64.391**	**0.977**
	DL^{2}F	70.406	0.965	77.323	0.963	66.431	0.975
	Δ	15.88%	−1.25%	19.79%	−1.09%	−3.17%	−0.23%
Humidity	MM5	10.353	0.680	10.158	0.685	9.982	0.697
	DL^{2}F	**7.484**	**0.817**	**7.778**	**0.818**	**7.61**	**0.813**
	Δ	−27.71%	20.15%	−23.43%	19.46%	−23.76%	16.67%

$$\sigma_n^2 = \frac{1}{N}\sum_{n=1}^{N}\left(r_n - \bar{r}\right)^2 \tag{11}$$

The statistical parameters are computed for both the MM5 and predicted databases, considering each climatic variable. Specifically, we compare the global solar radiation, atmospheric pressure, and relative humidity for the Casaccia and Ottana weather stations. Additionally, the temperature is plotted only for the Casaccia weather station due to the absence of measured values at the Ottana station. The generated Taylor diagrams, representing 1-day, 2-day, and 3-day predictions, are displayed from Fig. 9 to Fig. 11. In these diagrams, the “REF” point on the x-axis is the reference dataset, which consists of the measured values. Points that lie closest to the “REF” point indicate predicted or estimated patterns that agree well with the observations. The whole test sets are selected for the comparisons.

The charts clearly demonstrate the effectiveness of the proposed method in estimating meteorological parameters more accurately than MM5. The predicted values closely align with the measured trends, resulting in higher statistical variables for both weather stations.

In particular, for 1-day predictions (Fig. 9), the AI model better describes global solar radiation than standard MM5. The suggested approach exhibits a higher correlation coefficient, lower centred root-mean-square difference, and standard deviation for the analysed cities. Similar improvements can be observed for other meteorological parameters. For instance, temperatures (available only for Casaccia) are more precisely estimated, with an increase in correlation coefficient from 0.94 to 0.97, a decrease in centred root-mean-square difference from 0.35 to 0.23, and a decrement in standard deviation from 0.85 to 0.96.

The atmospheric pressure statistics are also improved for Casaccia, with an incremented correlation coefficient from 0.94 to 0.98 and a reduction in centred root-mean-square difference and standard deviation from 0.36 and 1.01 to 0.20 and 0.94, respectively. However, for Ottana, the estimations indicate a slightly poorer correlation coefficient and bigger RMSD than MM5, but the standard deviation is better.

The most significant improvement achieved with the AI model is observed for the relative humidity. The regional climate model (MM5) provides less accurate descriptions, as it lies farther from the REF point. On the contrary, the AI forecasting method increases the calculated correlation coefficient from 0.79 and 0.84 for Casaccia and Ottana, respectively, to 0.86 and 0.90. Moreover, the RMSD is reduced from 0.65 and 0.54 to 0.50 and 0.43, and the standard deviation is improved from 1.04 to 0.9 for Casaccia and from 0.94 to 0.91 for Ottana.

The trend observed in the 1-day predictions is consistently replicated in the 2-day and 3-day estimations, as shown in Figs. 10 and 11, respectively. The statistical parameters for both weather stations exhibit negligible variations across these prediction horizons. Once again, the suggested method proves to be effective in improving the accuracy of calculating climatic variables, resulting in better performance for solar radiation, temperature, and relative humidity.

Similar to the 1-day prediction, the atmospheric pressure in Ottana remains the only parameter that shows a different performance between the MM5 model and the AI forecasting approach for the 2-day and 3-day predictions. In this case, the MM5 model outperforms the AI method in terms of centred root-mean-square difference, with values of 0.15 and 0.14 for the 2-day and 3-day predictions, respectively, compared to 0.19 and 0.16 for the AI approach. On the other hand, the proposed AI method excels in terms of standard deviation, with values of 0.90 and 0.97 for the 2-day and 3-day predictions, respectively, compared to 1.00 for both estimations using the MM5 model.

Overall, the proposed model consistently improves the accuracy of meteorological predictions, providing better results for various climatic variables across different prediction horizons. However, slight variations in performance can be observed for specific parameters and locations, as evidenced by the atmospheric pressure in Ottana. Nonetheless, the proposed approach demonstrates significant potential in enhancing weather forecasting accuracy.

6. Worst case analysis

The primary objective of the present work was to ensure robust model performance under typical conditions, leading us to choose the MAE loss function over the MSE loss. While the latter tends to prioritise

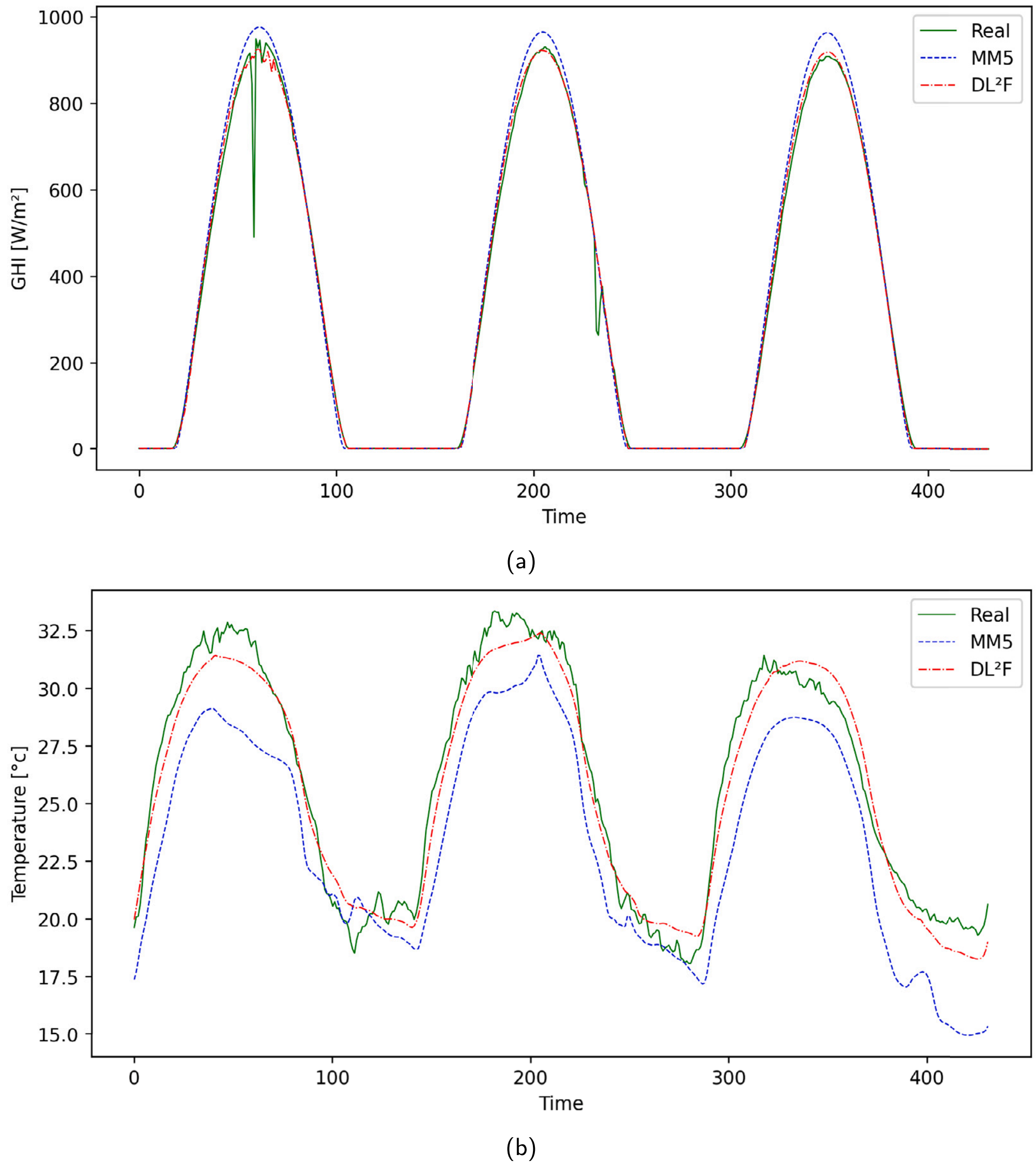


Fig. 6. Comparisons of GHI (a) and temperature (b) for Casaccia (3-day prediction).

minimising extreme errors observed in the presence of outliers, it may lead to poorer performance under standard conditions where errors are smaller but more frequent. We opted for the MAE loss to achieve a more balanced and consistent performance.

However, acknowledging the significance of evaluating model performance in non-standard scenarios with significant deviations in climatic variables, we comprehensively assessed such situations using MAE and R^2 metrics.

For the MAE metric, we identified the set of time steps where the absolute error between the values predicted by the MM5 system and the real values exceeded the third quartile. We applied the same procedure to the DL^{2}F system and considered the union of the two sets. This allowed us to focus on time steps where both systems exhibited suboptimal performance. Then, we computed and compared the MAE values of the two systems. A similar procedure was followed for the R^2 metric, considering squared errors between predicted and real values. More significant squared errors correspond to poorer (smaller) R^2 scores.

The choice to focus on errors greater than the third quartile aligned with insights from domain experts. Tables 5 and 6 report the results for Casaccia and Ottana, respectively.

In the case of Casaccia, we can observe that DL^{2}F consistently outperforms MM5 in terms of MAE metrics. When examining the R^2 metrics, comparable results are obtained for the variable GHI, but DL^{2}F shows worse results than MM5 for the Humidity variable.

In the case of Ottana, DL^{2}F outperforms MM5 in terms of MAE for the GHI and humidity variables. A behaviour similar to the standard case, where MM5 outperforms DL^{2}F, is observed for the atmospheric pressure. However, the predictions of both systems are pretty reliable, as indicated by the R^2 score exceeding 0.9 in both cases. Finally, DL^{2}F consistently outperforms MM5 regarding both metrics for the pressure parameter.

7. Deployment and scalability

This section provides preliminary insights into the deployment and scalability of the suggested solution in a production environment. They are drawn from additional experiments carried out with a version of DL^{2}F installed on our departmental cluster.

DL^{2}F trains a model for each location and meteorological variable once the time delay is fixed. Considering the small number of variables

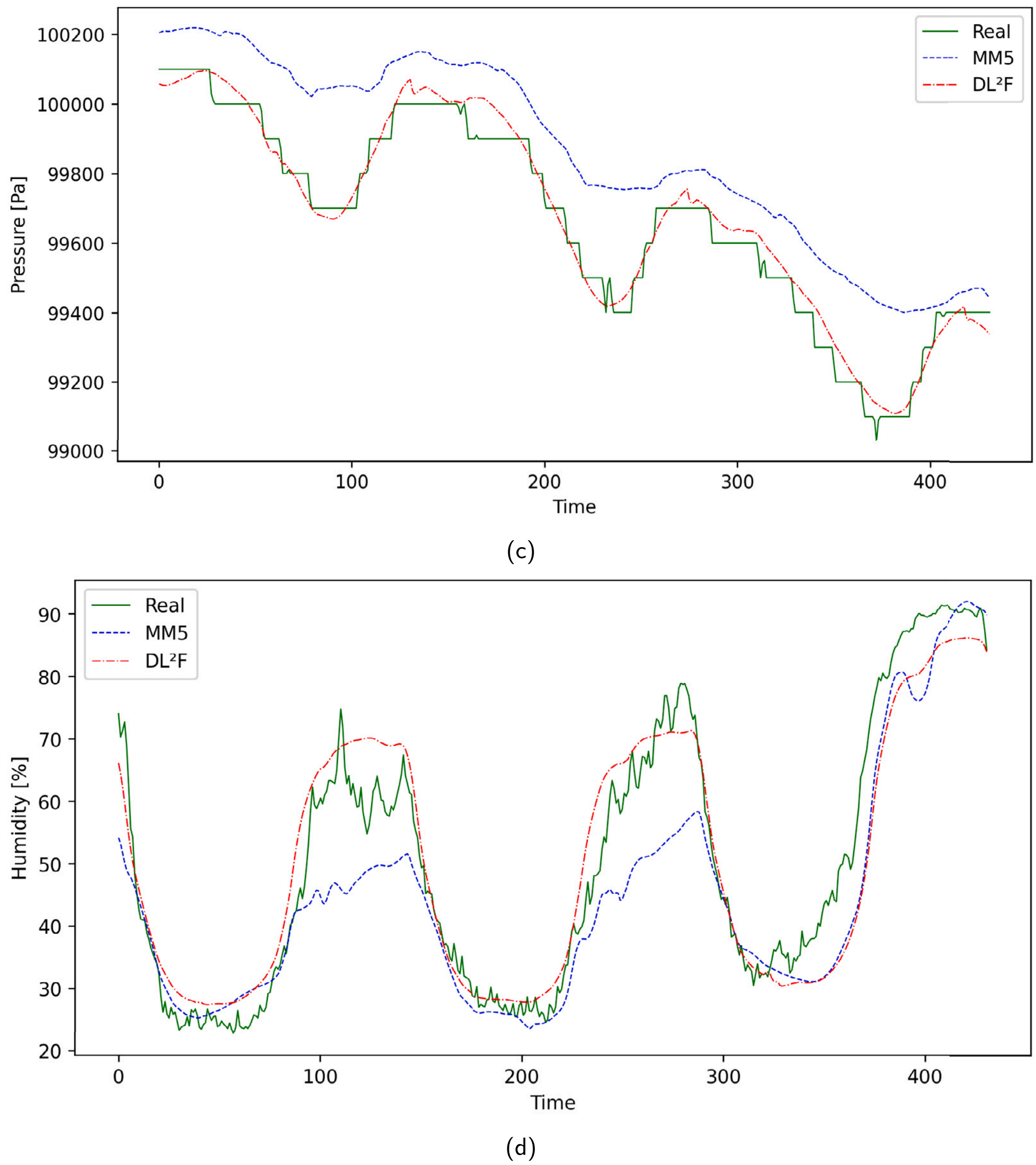
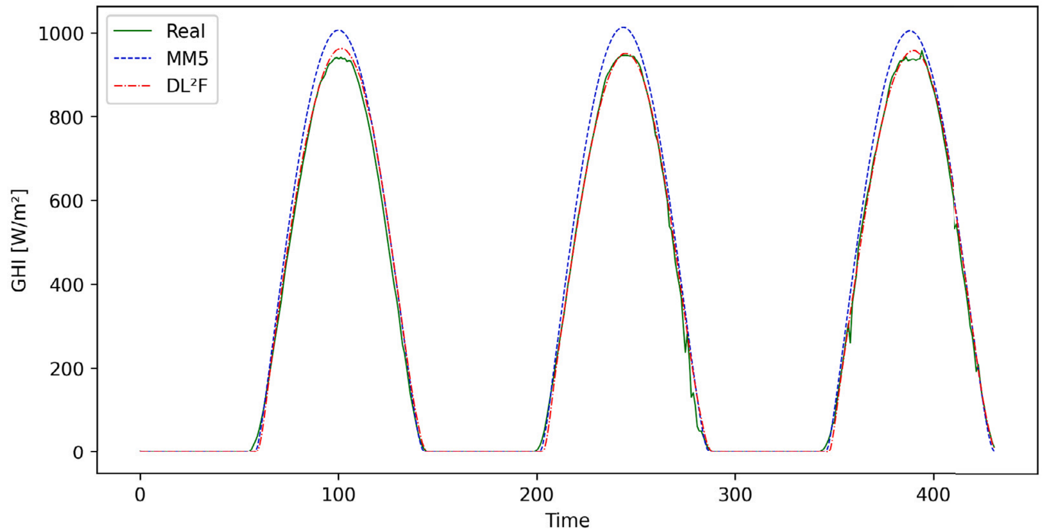


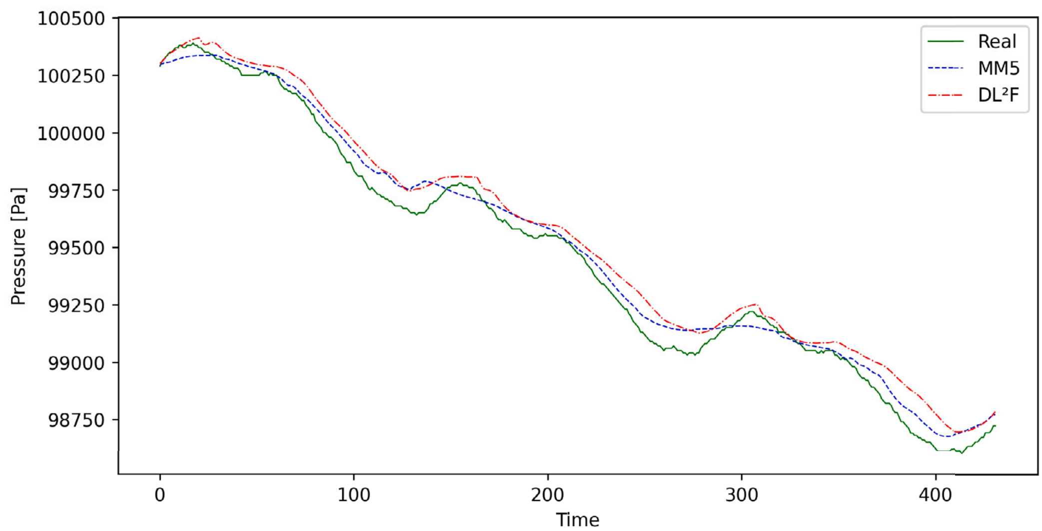
Fig. 7. Comparisons of atmospheric pressure (c) and relative humidity (d) for Casaccia (3-day prediction).

Table 5
Performance Metrics for Casaccia (values above the 3rd quartile).

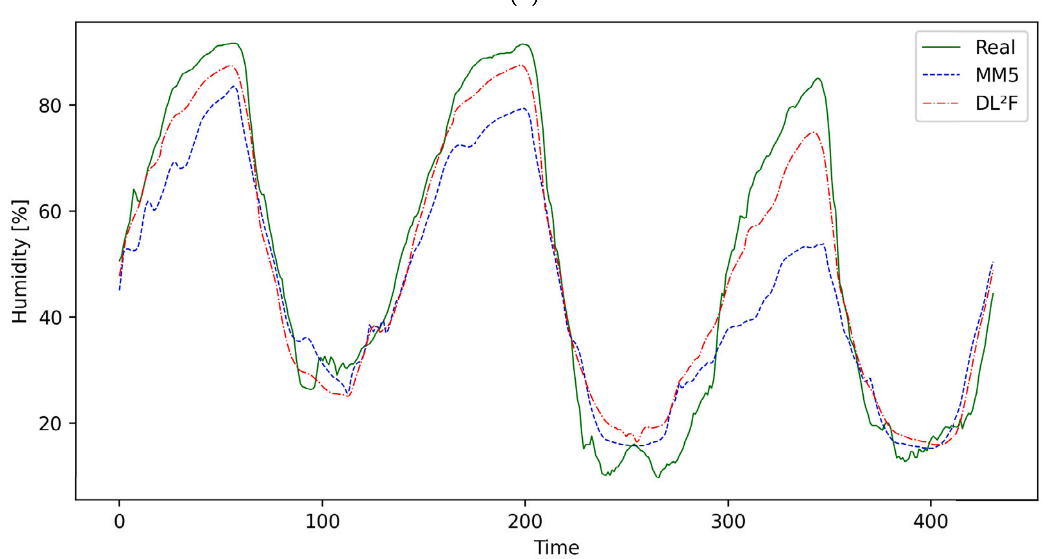
Variable	System	τ = 144 (1 day)		τ = 288 (2 days)		τ = 432 (3 days)	
		MAE	R^2	MAE	R^2	MAE	R^2
GHI	MM5	130.523	**0.666**	134.42	**0.644**	130.197	0.657
	DL^2F	**111.729**	0.664	**114.991**	0.638	**110.722**	**0.658**
	Δ	−14.4%	−0.21%	−14.45%	−0.94%	−14.96%	0.08%
Temperature	MM5	3.682	0.437	3.695	0.465	3.549	0.516
	DL^2F	**2.184**	**0.84**	**2.334**	**0.834**	**2.121**	**0.86**
	Δ	−40.68%	92.39%	−36.83%	79.57%	−40.24%	66.81%
Pressure	MM5	265.088	0.81	274.369	0.781	267.726	0.8
	DL^2F	**142.48**	**0.923**	**163.876**	**0.894**	**151.438**	**0.911**
	Δ	−46.25%	13.96%	−40.27%	14.43%	−43.44%	13.87%
Humidity	MM5	18.579	**−0.162**	18.76	**−0.171**	18.375	**−0.174**
	DL^2F	**14.536**	−0.22	**15.126**	−0.472	**14.691**	−0.333
	Δ	−21.76%	−35.49%	−19.37%	−175.48%	−20.05%	−91.31%



(a)



(b)



(c)

Fig. 8. Comparisons of GHI (a), atmospheric pressure (b) and relative humidity (c) for Ottana (3-day prediction).

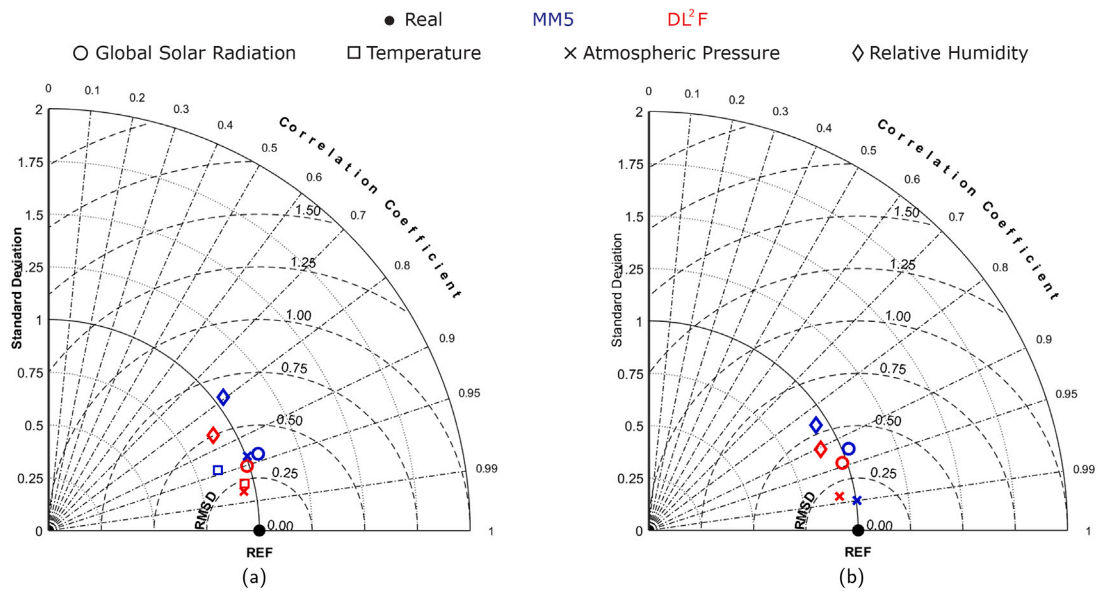


Fig. 9. Taylor diagrams for 1-day predictions referred to (a) Casaccia and (b) Ottana..

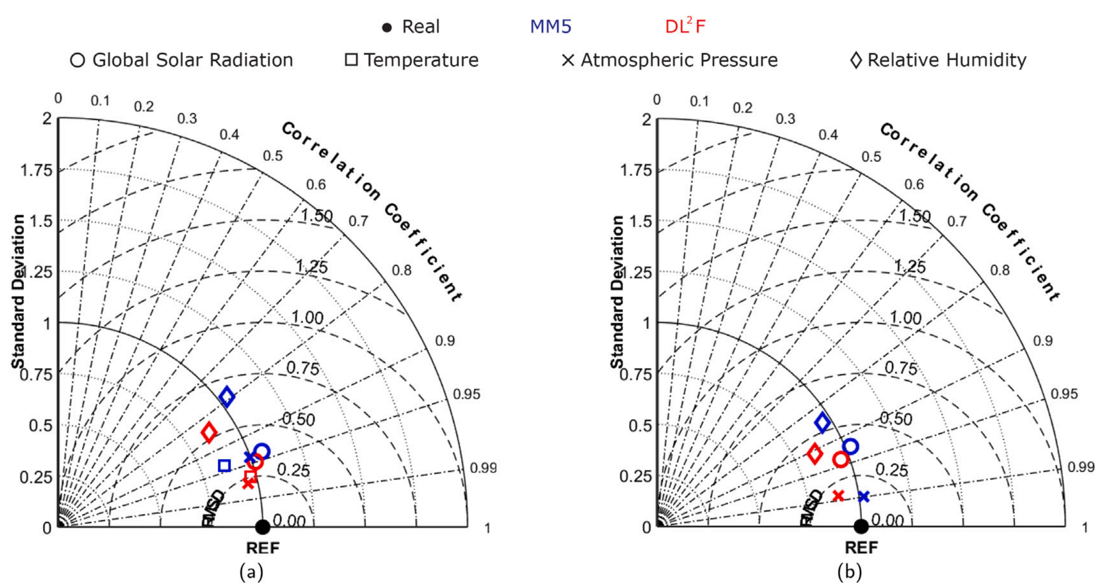


Fig. 10. Taylor diagrams for 2-day predictions referred to (a) Casaccia and (b) Ottana..

Table 6
Performance metrics for Ottana (values above the 3rd quartile).

Variable	System	$\tau = 144$ (1 day)		$\tau = 288$ (2 days)		$\tau = 432$ (3 days)	
		MAE	R^2	MAE	R^2	MAE	R^2
GHI	MM5	177.01	**0.433**	174.131	**0.465**	169.281	**0.452**
	DL^{2}F	**152.104**	0.428	**157.914**	0.41	**159.266**	0.405
	Δ	−14.07%	−1.25%	−9.31%	−11.77%	−5.92%	−10.23%
Pressure	MM5	**103.265**	**0.961**	**109.616**	**0.963**	**117.523**	**0.952**
	DL^{2}F	123.489	0.917	128.4	0.929	120.976	0.938
	Δ	19.58%	−4.57%	17.14%	−3.58%	2.94%	−1.56%
Humidity	MM5	16.778	−0.227	18.268	−0.417	18.638	−0.542
	DL^{2}F	**12.751**	**0.307**	**13.779**	**0.054**	**14.313**	**−0.007**
	Δ	−24.0%	235.28%	−24.57%	112.95%	−23.2%	98.62%

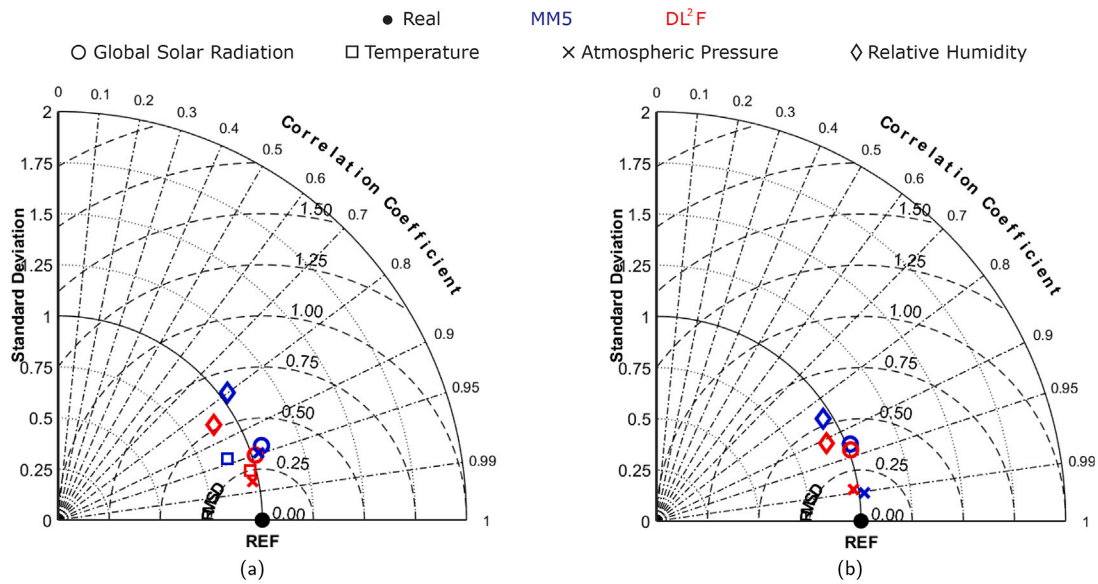


Fig. 11. Taylor diagrams for 3-day predictions referred to (a) Casaccia and (b) Ottana..

(only 4) and the limited set of locations with meteorological stations, scalability is manageable. The order of magnitude for potential locations is in the hundreds for each country; for instance, there are around 1000 in Italy. Moreover, the resource-intensive phase is the training process. Once trained, the model can efficiently make predictions using less high-performance hardware, eliminating the need for a GPU.

The system collects data from the previous day and the corresponding MM5 forecasts just after midnight. It then calculates the forecast for the next three days. Our system is deployed on a node of our departmental cluster, equipped with 2 *Intel Xeon 6-Core* processors and 32 GB of RAM, without any GPU. The time required to calculate forecasts for the two locations reported in the paper is less than 6 min. These observations highlight the model's efficiency and suitability for practical deployment in real-world scenarios.

8. Conclusions

We presented the first version of DL^{2}F, a GRU-based deep learning model designed to enhance the forecasts returned by the MM5 system for four weather variables: global horizontal irradiance, temperature, atmospheric pressure, and relative humidity.

The model inputs a time series for each variable, along with the prediction returned by the MM5 system. Then, it generates a new forecast for every variable at the same future time.

The experiments conducted on two case studies demonstrate that the proposed model consistently improves the MM5 system's accuracy, effectively addressing its tendency to overestimate real data in some seasons. It is worth noting that the proposed model was developed in geographical locations where historical real data for the four weather variables are available, i.e., locations where a weather station capable of providing relevant measurements is present.

In future works, attention mechanisms will be explored and integrated into the RNN to increase the interpretability of the here-suggested AI approach. Moreover, we plan to extend the presented approach to include sites where measured data are inaccessible. We also aim to expand the analyses and include variables with more dynamic and noisy trends, such as wind speed and direction, using meteorological masts devoted to wind farm site assessment. By doing so, we strive to increase further the accuracy of forecasts provided by the MM5 system.

CRediT authorship contribution statement

Luciano Caroprese: Conceptualization, Data curation, Formal analysis, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing, Investigation. **Mariano Pierantozzi:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Camilla Lops:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Sergio Montelpare:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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